N O V A P L E X

Cinema System

Project Documentation

Object Oriented Software Development

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Module	Object Oriented Software Development

1. System Description

The NovaPlex Cinema Management System is a Java desktop app made to make it easier for admins of the NovaPlex cinemas to create, retrieve, update and delete data. It was made with accessibility and user friendliness in mind so that the UI can be easily navigated.

The system connects to a MySQL database using JDBC (Java Database Connectivity) and provides a fully featured graphical user interface built using Java Swing. The interface allows administrators to manage all data through a tabbed panel layout, with each tab dedicated to a specific area of the cinema's operations.

Access to the system is restricted to administrators through a secure login screen that validates credentials against the database using BINARY case sensitive password comparison, and checks that the user is both active and has Admin level permissions.

The system has data validation and logic including duplicate prevention, foreign key validation, soft deletes for movies and users, prevention of double booking seats, prevention of booking broken seats, and prevention of creating showtimes for retired movies.

1.1 Technology Stack

Technology	Details
Language	Java (JDK 17+)
GUI Framework	Java Swing
Database	MySQL 8.0
DB Connector	mysql-connector-j-9.6.0.jar
IDE	Visual Studio Code

1.2 Tabs

- Movies — Add, update, retire and view all movies including genre, duration, age rating and status
- Showtimes — Schedule movie screenings with automatic end time calculation based on movie duration
- Bookings — Manage customer bookings for both registered users and guests
- Screens — Manage cinema screens and their seating capacity
- Seats — Configure seating layouts including VIP and Standard seats
- Users — Manage registered customer and staff accounts with role-based access
- Payments — Track and manage payment transactions
- Tickets — Generate and manage digital tickets
- Seat Bookings — Link specific seats to bookings for each showtime
- Seat Reserves — Temporarily reserve seats with automatic 60sec check for expiry
- Reports — Generate sales reports with filtering by booking status showing total revenue and booking counts

2. System Requirements

The following user stories were identified during the requirements gathering phase of the project.

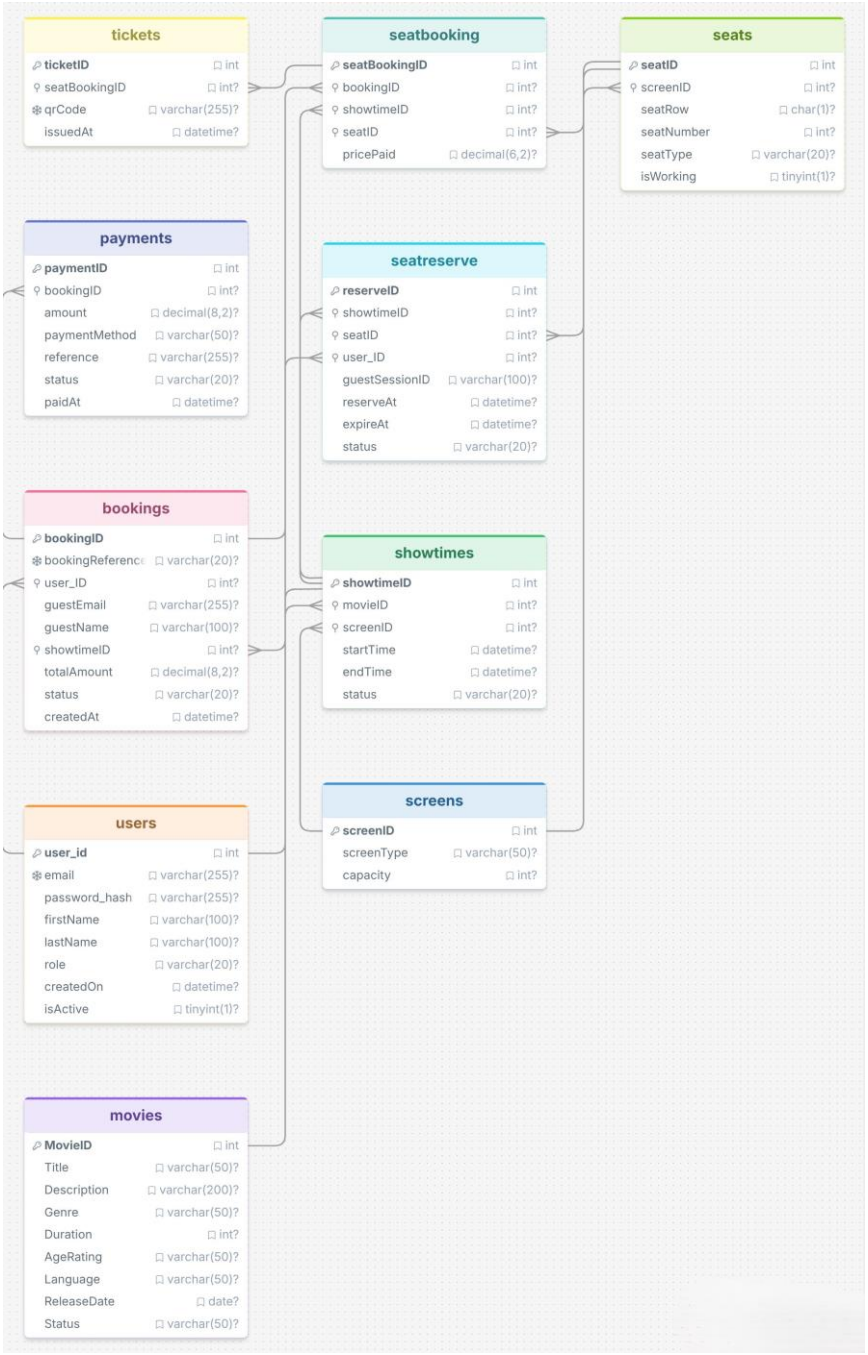
2.1 Admin User Stories

ID	Role	User Story	Status
ADMIN-001	Admin	As an admin I want to manage screens and seats so I can configure the cinema layout	**Implemented**
ADMIN-002	Admin	As an admin I want to generate sales reports so I can see total revenue and booking counts	**Implemented**
ADMIN-003	Admin	As an admin I want to manage movies so I can add, update and retire them	**Implemented**
ADMIN-004	Admin	As an admin I want to manage showtimes so I can schedule movie screenings	**Implemented**
ADMIN-005	Admin	As an admin I want to manage users so I can add, update and deactivate accounts	**Implemented**
ADMIN-006	Admin	As an admin I want to log in securely so only authorised staff can access the system	**Implemented**
ADMIN-007	Admin	As an admin I want to manage bookings so I can add, update and cancel reservations	**Implemented**
ADMIN-008	Admin	As an admin I want to manage payments so I can track transactions	**Implemented**
ADMIN-009	Admin	As an admin I want seat reserves to auto expire after 15 minutes	**Implemented**

3. Database Design

The NovaPlex system uses a MySQL relational database named novaplexdb. The database consists of 10 tables with foreign key relationships between them to maintain data integrity.

3.1 ER Diagram



3.2 Database Table Structures and Data

The following screenshots show the structure and data of the key tables in the novaplexdb database.

3.2.1 Movies Table — Structure

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
MovieID	int	NO	PRI	NULL	auto_increment
Title	varchar(50)	YES		NULL	
Description	varchar(200)	YES		NULL	
Genre	varchar(50)	YES		NULL	
Duration	int	YES		NULL	
AgeRating	varchar(50)	YES		NULL	
Language	varchar(50)	YES		NULL	
ReleaseDate	date	YES		NULL	
Status	varchar(50)	YES		NULL	

3.2.2 Movies Table — Data

MovieID	Title	Description	Genre	Duration	AgeRating	Language	ReleaseDate	Status
2	Brother Bear	A beautiful story about two brothers	Family	85	PG	English	2003-12-05	Showing
3	Lion King	A young lion who has to embrace his destiny to ...	Family	88	PG	English	1994-06-24	Inactive
4	The Dark Knight	When the menace known as the Joker wreaks h...	Action	152	12A	English	2008-07-18	Now Showing
5	Inception	A thief who steals corporate secrets through th...	Sci-Fi	148	12A	English	2010-07-16	Now Showing
6	Interstellar	A team of explorers travel through a wormhole i...	Sci-Fi	169	12A	English	2014-11-07	Inactive
7	The Shawshank Redemption	Two imprisoned men bond over a number of ye...	Drama	142	15	English	1994-09-23	Now Showing
8	Oppenheimer	The story of American scientist J. Robert Oppe...	Drama	180	15	English	2023-07-21	Inactive
9	The Super Mario Bros Movie	A plumber named Mario travels through an und...	Animation	92	PG	English	2023-04-05	Now Showing
10	Avengers Endgame	After the devastating events of Infinity War, th...	Action	181	12A	English	2019-04-26	Now Showing
11	Dune Part Two	Paul Atreides unites with Chani and the Fremen ...	Sci-Fi	166	12A	English	2024-03-01	Inactive
12	Inside Out 2	Joy, Sadness, Anger, Fear and Disgust navigat...	Animation	100	PG	English	2024-06-14	Inactive
13	Deadpool and Wolverine	Deadpool is offered a chance to join the Time V...	Action	127	15	English	2024-07-26	Coming Soon
14	Wicked	The untold story of the witches of Oz. Elphaba ...	Romance	160	PG	English	2024-11-22	Coming Soon
15	Moana 2	Moana receives an unexpected call from her wa...	Animation	100	PG	English	2024-11-27	Coming Soon
16	Lion King	A young lion who has to embrace his destiny to ...	Family	88	PG	English	1994-06-24	Now Showing
17	King Charles	A movie about a king	Thriller	60	12A	English	2000-10-05	Coming Soon
18	Hamilton	A shakespear play into a movie	Drama	62	12A	English	2005-09-15	Coming Soon
19	Moana	A lovely story about a young girl	Animation	65	PG	English	2005-08-09	Coming Soon
20	Lion King	A young lion who has to embrace his destiny to ...	Family	88	PG	English	1994-06-24	Inactive
21	Aladin	A rif raf who wants to be a prince	Animation	78	PG	English	2005-09-08	Coming Soon
22	Tinker Bell	A fairy	Animation	54	PG	English	2005-09-08	Coming Soon
23	Turning Red	A girl turns into a red panda	Animation	78	PG	English	2026-04-04	Now Showing
24	Frozen	A girl with ice powers	Animation	78	PG	English	2026-05-04	Coming Soon

3.2.3 Showtimes Table — Structure

Field	Type	Null	Key	Default	Extra
showtimeID	int	NO	PRI	NULL	auto_increment
movieID	int	YES	MUL	NULL	
screenID	int	YES	MUL	NULL	
startTime	datetime	YES		NULL	
endTime	datetime	YES		NULL	
status	varchar(20)	YES		NULL	

3.2.4 Showtimes Table — Data

showtimeID	movieID	screenID	startTime	endTime	status
1	3	1	2026-09-23 19:10:00	2026-09-23 20:38:00	Scheduled
2	2	2	2026-09-23 19:10:00	2026-09-23 20:35:00	Scheduled
3	2	1	2026-04-25 14:00:00	2026-04-25 15:25:00	Cancelled
4	3	2	2026-04-25 17:00:00	2026-04-25 18:28:00	Scheduled
5	4	1	2026-04-25 19:00:00	2026-04-25 21:32:00	Scheduled
6	5	2	2026-04-26 14:00:00	2026-04-26 16:28:00	Scheduled
7	6	1	2026-04-26 18:00:00	2026-04-26 20:49:00	Scheduled
8	14	2	2026-10-13 15:15:00	2026-10-13 17:55:00	Scheduled
9	7	2	2026-10-04 15:10:00	2026-10-04 17:32:00	Cancelled
10	13	2	2026-08-12 15:10:00	2026-08-12 17:17:00	Scheduled
11	12	2	2026-08-09 14:00:00	2026-08-09 15:40:00	Cancelled
12	19	2	2026-05-11 15:00:00	2026-05-11 16:05:00	Cancelled
13	13	1	2026-05-11 15:00:00	2026-05-11 17:07:00	Scheduled
14	10	2	2026-09-08 15:00:00	2026-09-08 18:01:00	Scheduled

3.2.5 Bookings Table — Structure

Field	Type	Null	Key	Default	Extra
bookingID	int	NO	PRI	NULL	auto_increment
bookingReference	varchar(20)	YES	UNI	NULL	
user_ID	int	YES	MUL	NULL	
guestEmail	varchar(255)	YES		NULL	
guestName	varchar(100)	YES		NULL	
showtimeID	int	YES	MUL	NULL	
totalAmount	decimal(8,2)	YES		NULL	
status	varchar(20)	YES		NULL	
createdAt	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

3.2.6 Bookings Table — Data

bookingID	bookingReference	user_ID	guestEmail	guestName	showtimeID	totalAmount	status	createdAt
1	NOVA-001	NULL	Charles@gmail.com	Charlie	1	25.00	Cancelled	2026-04-25 01:46:03
12	NP-002	NULL	sarah.connor@email.com	Sarah Connor	3	40.00	Confirmed	2026-04-25 01:46:03
13	NP-003	NULL	tom.ryan@email.com	Tom Ryan	4	50.00	Cancelled	2026-04-25 01:46:03
14	NP-004	NULL	emma.walsh@email.com	Emma Walsh	4	58.00	Confirmed	2026-04-25 01:46:03
15	NP-005	NULL	liam.kelly@email.com	Liam Kelly	5	50.00	Cancelled	2026-04-25 01:46:03
16	NP-006	NULL	aoife.brennan@email.com	Aoife Brennan	5	12.00	Confirmed	2026-04-25 01:46:03
17	NP-USER-001	1	NULL	NULL	3	24.00	Confirmed	2026-04-25 01:46:03
18	NP-USER-002	4	NULL	NULL	3	40.00	Confirmed	2026-04-25 01:44:09

3.2.7 Users Table — Structure

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
email	varchar(255)	YES	UNI	NULL	
password_hash	varchar(255)	YES		NULL	
firstName	varchar(100)	YES		NULL	
lastName	varchar(100)	YES		NULL	
role	varchar(20)	YES		NULL	
createdOn	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
isActive	tinyint(1)	YES		NULL	

3.2.8 Users Table — Data

user_id	email	password_hash	firstName	lastName	role	createdOn	isActive
1	sarahcrotty@novaplex.com	sarah001	Sarah	Crotty	Admin	2026-04-25 01:46:03	1
3	john.murphy@email.com	pass123	John	Murphy	Customer	2026-04-25 01:46:03	1
4	sarah.connor@email.com	pass123	Sarah	Connor	Customer	2026-04-25 01:46:03	1
5	tom.ryan@email.com	pass123	Tom	Ryan	Customer	2026-04-25 01:46:03	0
6	staff@novaplex.com	staff123	Cinema	Staff	Staff	2026-04-25 01:46:03	1
7	manager@novaplex.com	manager123	Cinema	Manager	Admin	2026-04-25 01:46:03	0
8	gordanramsey@gmail.com	gordan	gordan	ramsey	Customer	2026-04-25 01:36:22	1

3.3 Key Relationships

- showtimes references movies (movieID) and screens (screenID)
- bookings references showtimes (showtimeID) and optionally users (user_ID)
- seats references screens (screenID)
- seatbooking references bookings, showtimes and seats
- seatreserve references showtimes, seats and optionally users
- payments references bookings (bookingID)
- tickets references seatbooking (seatBookingID)

4. GUI Design

The graphical user interface was built using Java Swing. The application launches with a secure login screen and upon successful login it opens the main tabbed interface. The design uses a consistent dark navy colour scheme throughout.

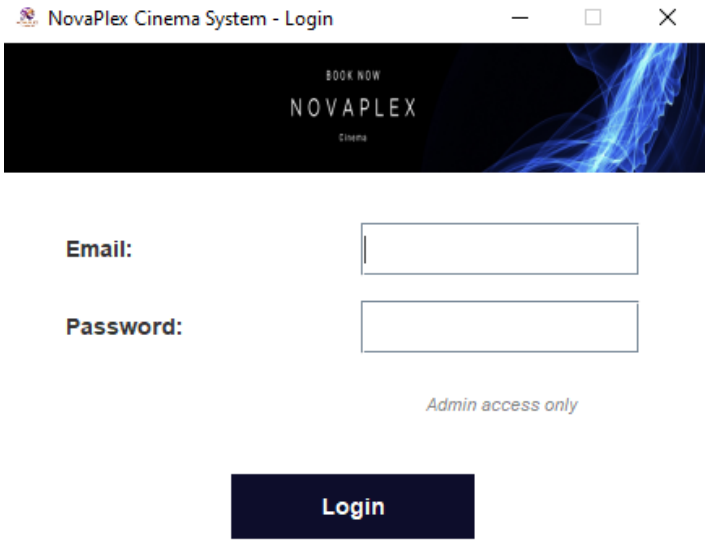
4.1 Swing Components Used

Component	Usage
JFrame	Main application window and login window
JTabbedPane	Main navigation between 11 management panels
JTable	Display all database records with alternating row colours
JScrollPane	Scrollable table views
JButton	Add, Update, Delete and Refresh actions with colour coding
JTextField	Text input for all data entry fields
JPasswordField	Secure password input in login and user panels
JComboBox	Dropdown menus for status, genre, role and type fields
JLabel	Field labels and banner display
JOptionPane	Success, error and confirmation dialogs
Timer	Auto expire seat reservations every 15 minutes

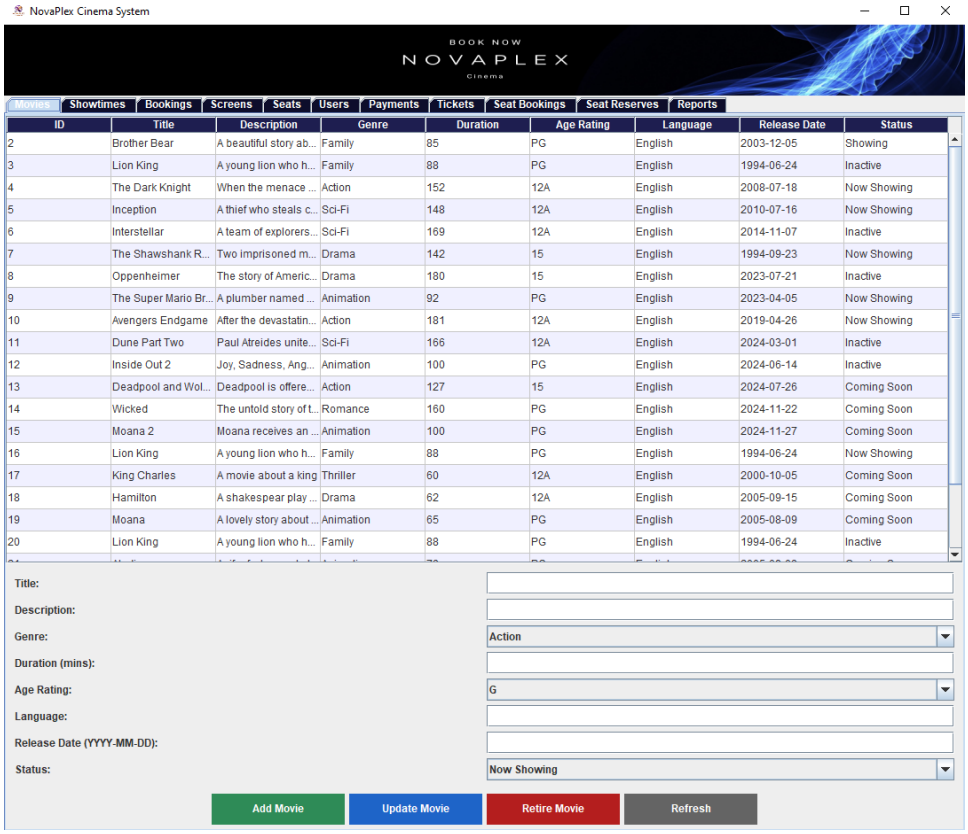
4.2 Colour Scheme

Element	Colour	RGB Value
Background / Header	Dark Navy	(13, 13, 43)
Add Button	Green	(46, 139, 87)
Update Button	Blue	(30, 100, 200)
Delete / Retire Button	Red	(180, 30, 30)
Refresh Button	Grey	(100, 100, 100)

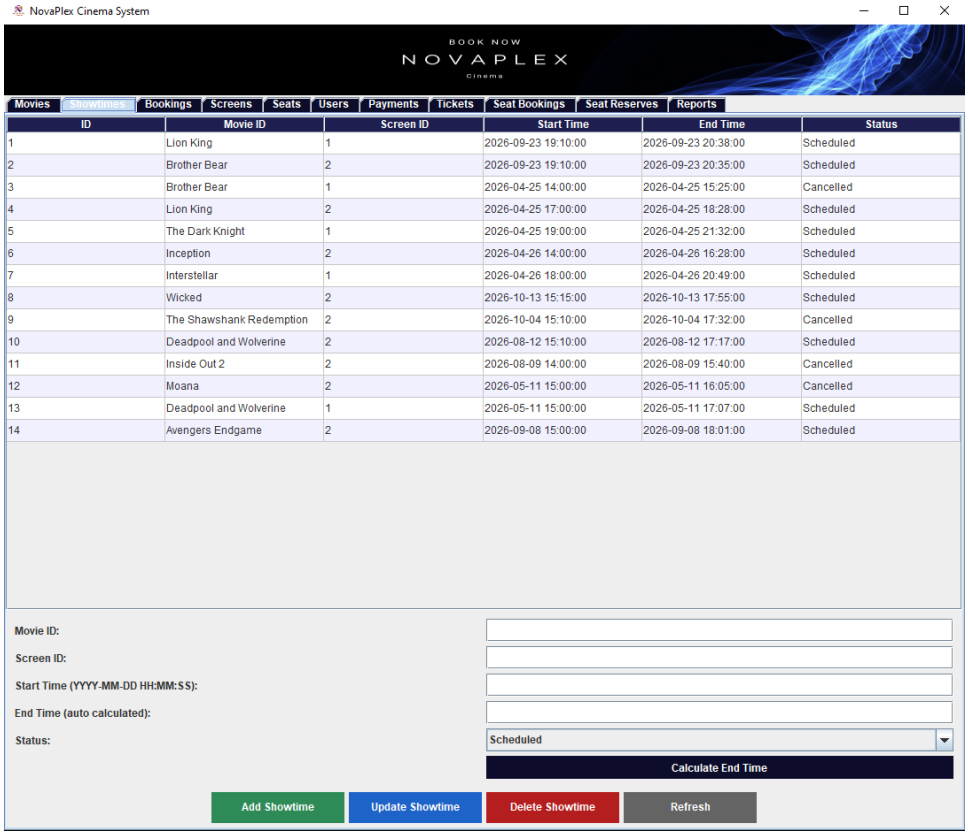
4.3 Login Screen GUI



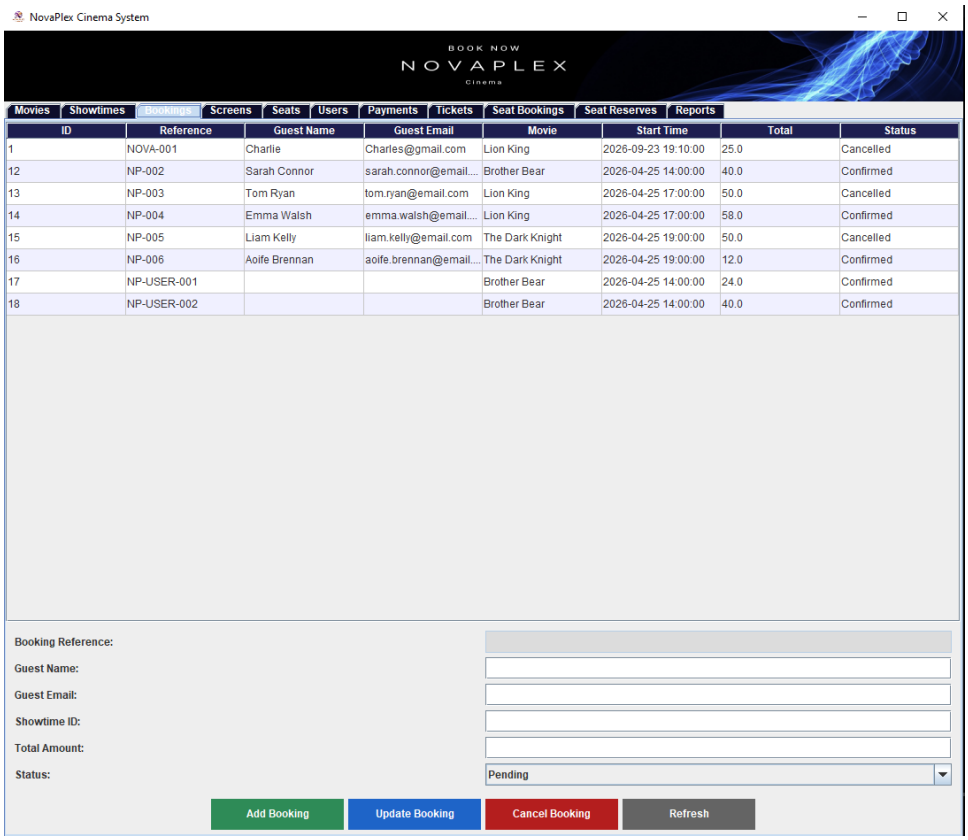
4.4 Movies GUI



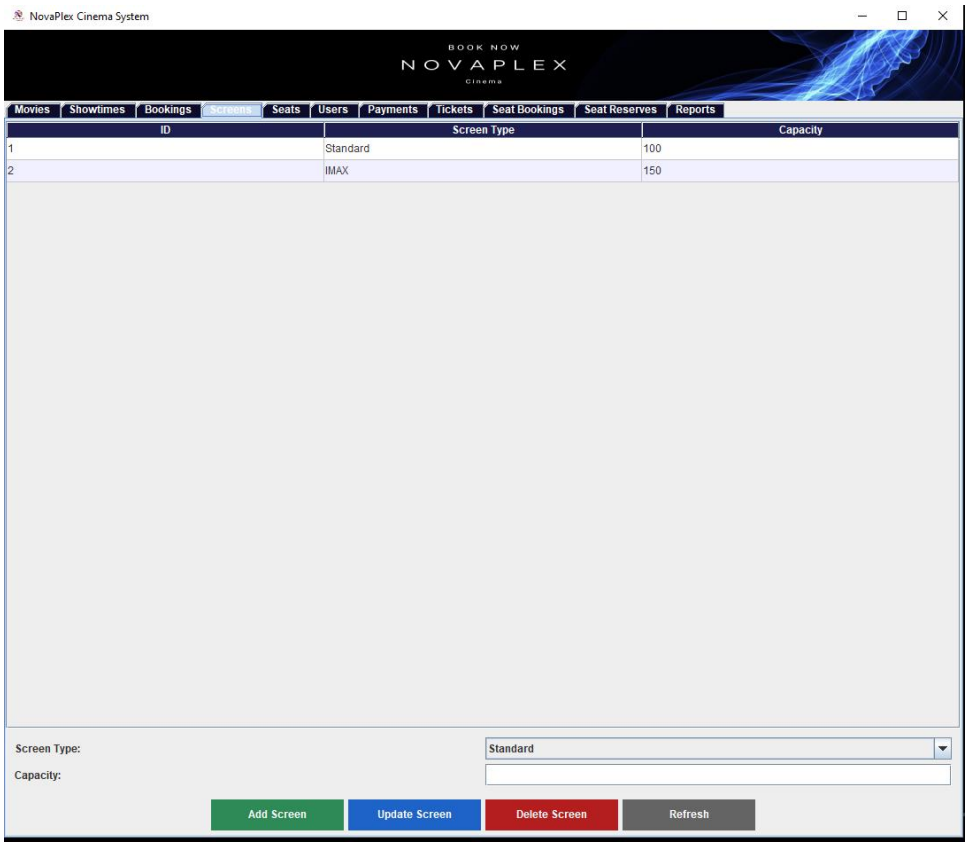
4.5 Showtimes GUI



4.6 Bookings GUI



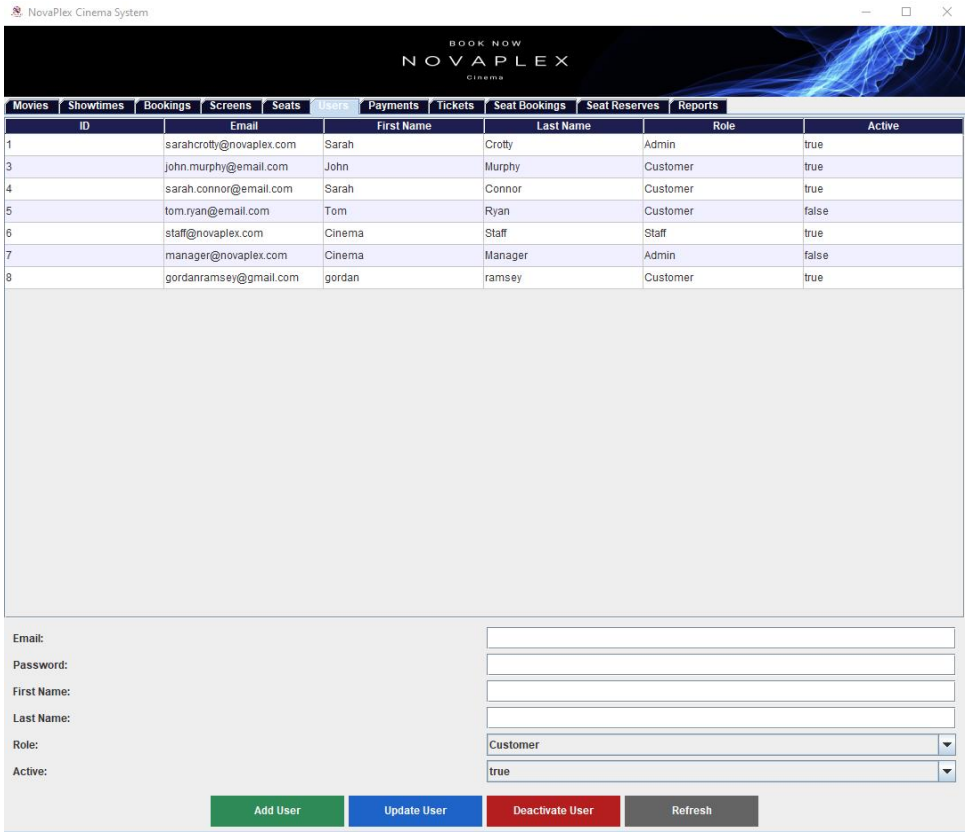
4.7 Screens GUI



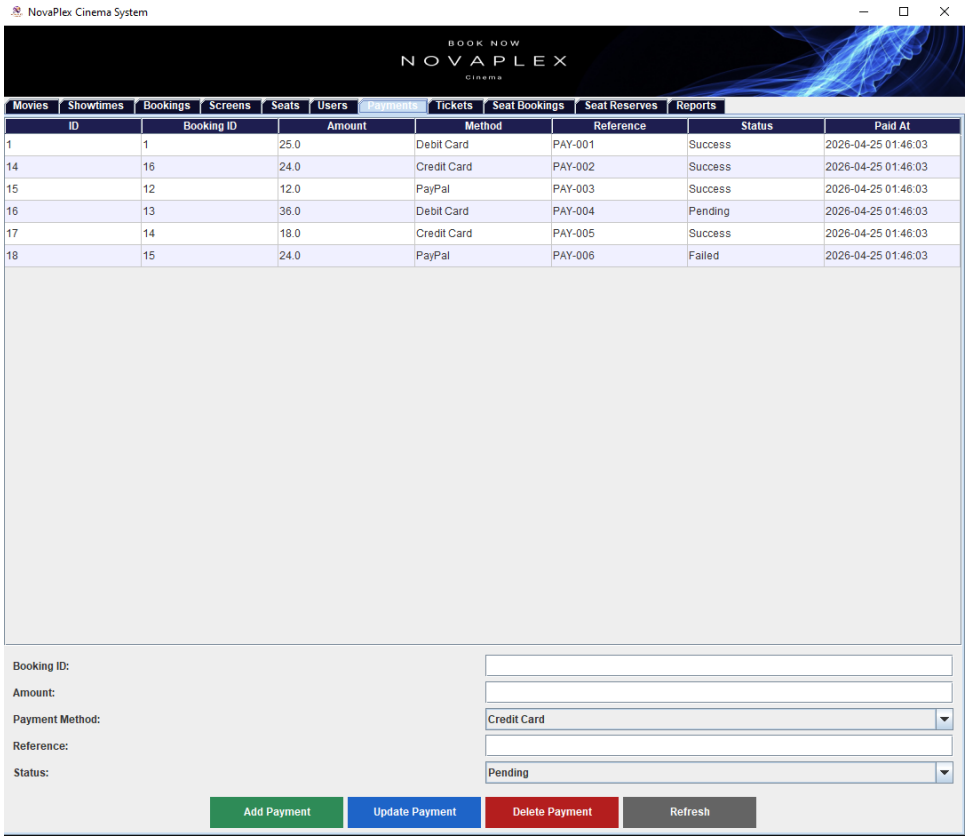
4.8 Seats GUI



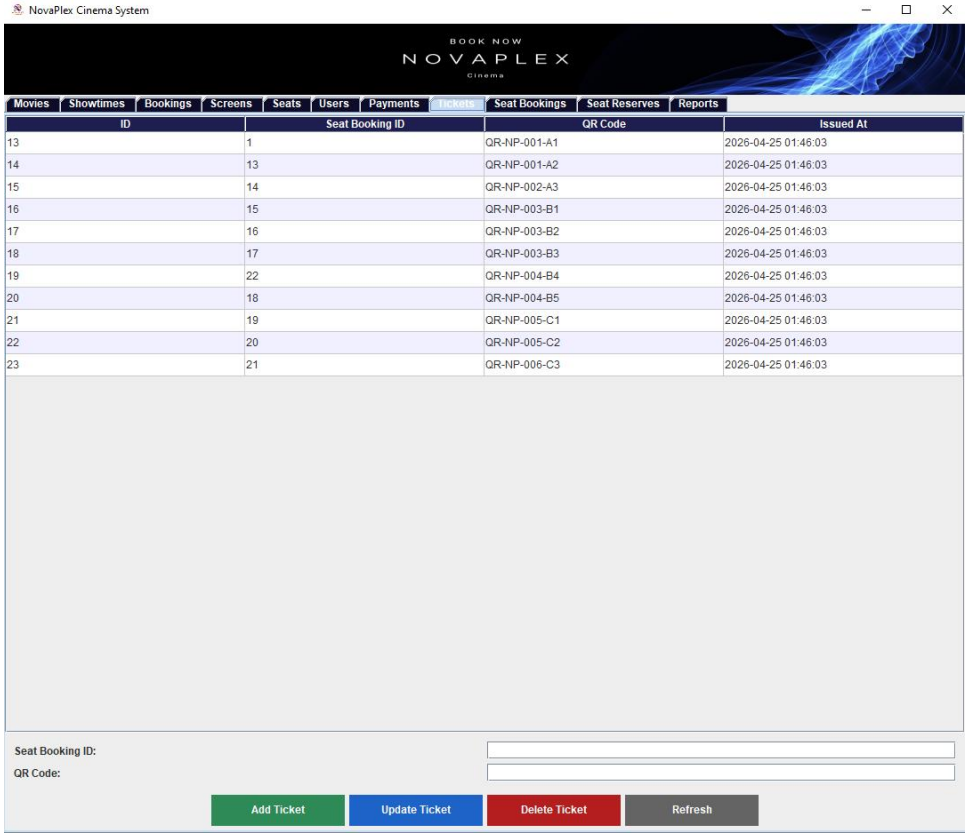
4.9 Users GUI



4.10 Payments



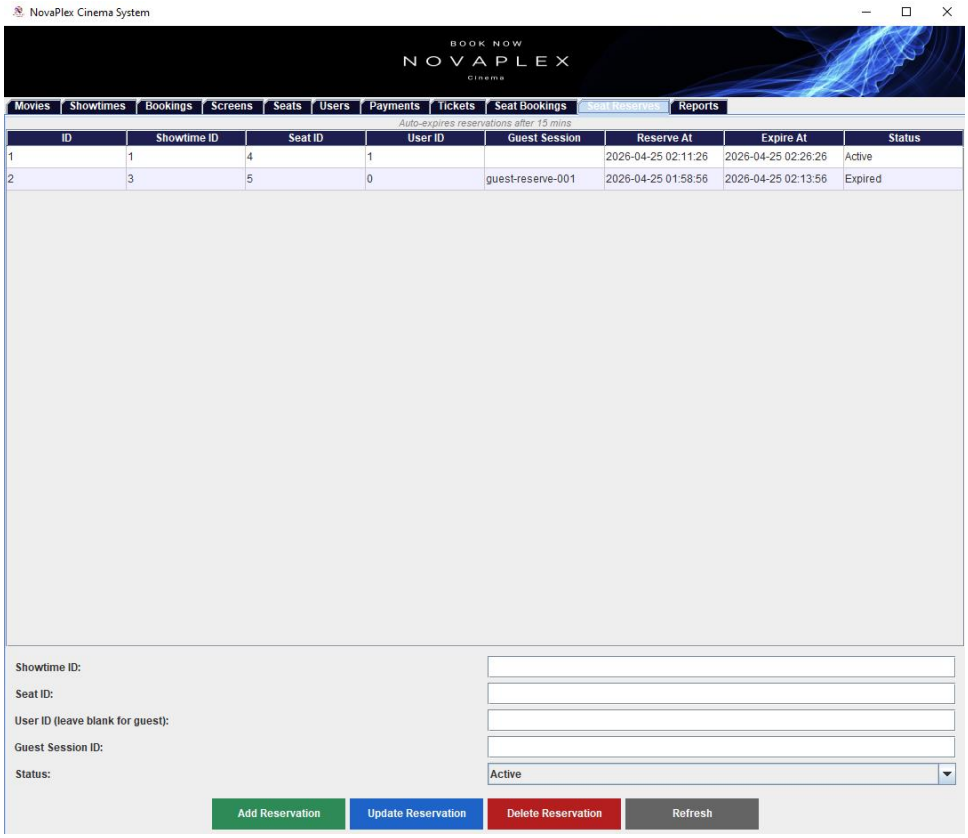
4.12 Tickets GUI



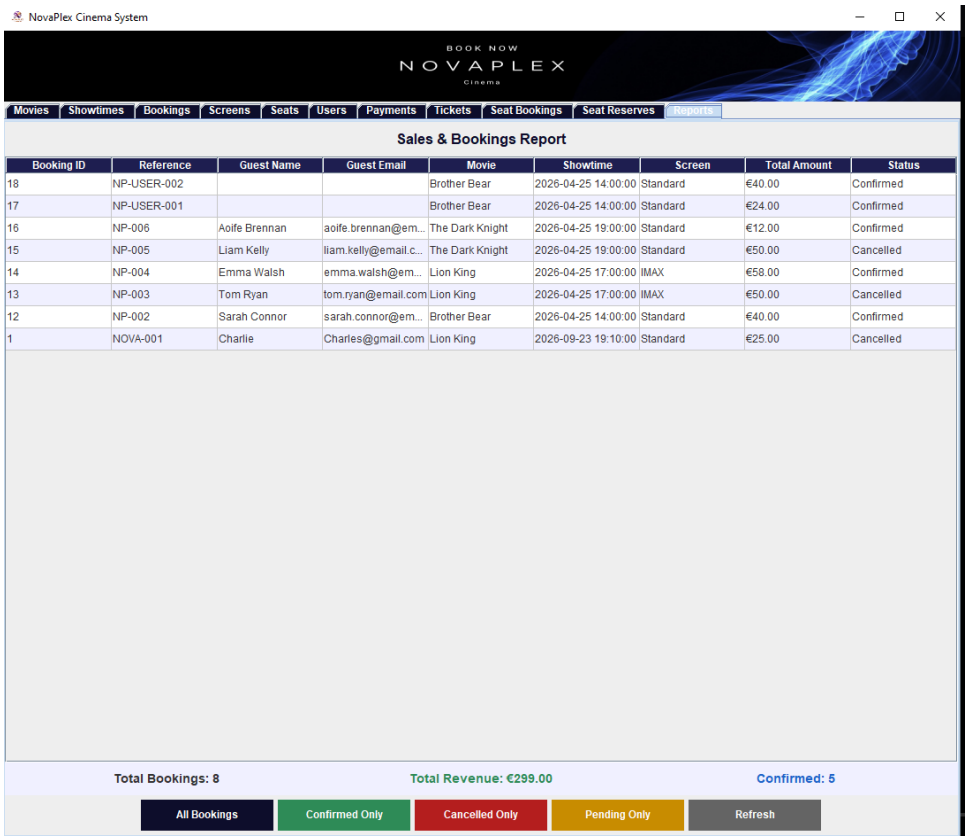
4.13 Seat Bookings GUI



4.14 Seat Reserves GUI



4.15 Reports GUI



5. Interesting Code Snippets

The following code snippets highlight the most interesting technical implementations in the NovaPlex system.

5.1 Auto Calculate End Time (ShowtimePanel.java)

When adding a showtime, the end time is automatically calculated by fetching the movie duration from the database and adding it to the start time using Java's LocalDateTime API. This prevents human error when scheduling screenings.

```
int duration = GUIHelper.getMovieDuration(movieID);
java.time.LocalDateTime startTime = java.time.LocalDateTime.parse(start,
    java.time.format.DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm:ss"));
java.time.LocalDateTime endTime = startTime.plusMinutes(duration);
String endFormatted = endTime.format(
    java.time.format.DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm:ss"));
endField.setText(endFormatted);
```

5.2 Case Sensitive Admin Login (LoginGUI.java)

The login system uses MySQL's BINARY keyword to enforce case sensitive password comparison. It also checks that the user has the Admin role and is currently active before granting access.

```
String sql = "SELECT * FROM users WHERE email=? " +
    "AND BINARY password_hash=? " +
    "AND role='Admin' AND isActive=1";
```

5.3 Soft Delete for Movies (MovieDAO.java)

Rather than permanently deleting movies from the database, the system performs a soft delete by setting the status to Inactive. This preserves historical data and prevents orphaned records in related tables.

```
String sql = "UPDATE movies SET status='Inactive' WHERE movieID=?";
```

5.4 Soft Delete for Users (UserDAO.java)

Similarly, users are never permanently deleted. Their isActive flag is set to 0 which prevents them from logging in while preserving their booking history.

```
String sql = "UPDATE users SET isActive=0 WHERE user_id=?";
```

5.5 Prevent Double Booking (GUIHelper.java)

Before allowing a seat booking the system checks if that seat is already booked for that showtime. Cancelled bookings are excluded so their seats can be rebooked.

```
String sql = "SELECT sb.seatBookingID FROM seatbooking sb " +
    "INNER JOIN bookings b ON sb.bookingID = b.bookingID " +
    "WHERE sb.seatID=? AND sb.showtimeID=? " +
    "AND b.status != 'Cancelled'";
```

5.6 Sales Report with 4-Table INNER JOIN (ReportPanel.java)

The reports panel uses an INNER JOIN across four tables to produce a comprehensive sales report showing booking details alongside movie titles and screen types.

```
String sql = "SELECT b.bookingID, b.bookingReference, b.guestName, " +
    "b.guestEmail, m.title, s.startTime, sc.screenType, " +
    "b.totalAmount, b.status " +
    "FROM bookings b " +
    "INNER JOIN showtimes s ON b.showtimeID = s.showtimeID " +
    "INNER JOIN movies m ON s.movieID = m.movieID " +
    "INNER JOIN screens sc ON s.screenID = sc.screenID";
```

5.7 Auto Expire Seat Reservations (MainGUI.java + SeatReserveDAO.java)

A javax.swing.Timer runs in the background every 60 seconds and automatically expires seat reservations older than 15 minutes.

```
javax.swing.Timer autoExpireTimer = new javax.swing.Timer(60000,
    new ActionListener() {
        public void actionPerformed(ActionEvent e) {
            new SeatReserveDAO().expireOldReservations();
        }
    });
autoExpireTimer.start();
// SQL in SeatReserveDAO.java:
String sql = "UPDATE seatreserve SET status='Expired' " +
    "WHERE status='Active' " +
    "AND reserveAt < DATE_SUB(NOW(), INTERVAL 15 MINUTE)";
```

5.8 Foreign Key Validation Before Insert (GUIHelper.java)

Before any insert or update operation the system validates that all referenced IDs exist in the database. The same pattern is used for movies, screens, showtimes, seats, bookings and users.

```
public static boolean movieExists(int movieID) {
    String sql = "SELECT movieID FROM movies WHERE movieID=?";
    PreparedStatement ps = con.prepareStatement(sql);
    ps.setInt(1, movieID);
    ResultSet rs = ps.executeQuery();
    return rs.next();
}
```

6. Test Cases

The following test cases were carried out to verify the system works correctly across all panels.

6.1 Login Tests

Test	Expected Result	Pass/Fail
Login with valid admin credentials	Login successful, main GUI opens	**Pass**
Login with wrong password	Error message shown	**Pass**
Login with wrong case password	Error message case sensitive check	**Pass**
Login with deactivated admin account	Error message shown	**Pass**
Login with non-admin user	Error message shown	**Pass**

6.2 Movie Tests

Test	Expected Result	Pass/Fail
Add movie with valid data	Movie added successfully	**Pass**
Add movie with empty fields	Error message shown	**Pass**
Add movie with invalid date format	Error message shown	**Pass**
Add duplicate movie title and release date	Error message shown	**Pass**
Retire movie with no upcoming showtimes	Status changes to Inactive	**Pass**
Retire movie with upcoming showtimes	Error message shown	**Pass**
Update movie with no changes made	Error message shown	**Pass**

6.3 Showtime Tests

Test	Expected Result	Pass/Fail
Calculate end time with valid movie and start time	End time auto calculated correctly	**Pass**
Add showtime for retired movie	Error message shown	**Pass**
Add showtime with end before start	Error message shown	**Pass**

Add showtime with invalid movie ID	Error message shown	**Pass**
Add duplicate showtime	Error message shown	**Pass**
Delete showtime with active bookings	Error message shown	**Pass**

6.4 Booking Tests

Test	Expected Result	Pass/Fail
Add booking with valid data	Booking added successfully	**Pass**
Add booking with invalid email format	Error message shown	**Pass**
Add booking with total amount of 0	Error message shown	**Pass**
Add duplicate booking reference	Error message shown	**Pass**
Cancel a booking	Status changes to Cancelled	**Pass**

6.5 Seat Booking Tests

Test	Expected Result	Pass/Fail
Book an available seat	Seat booked successfully	**Pass**
Book a seat already booked for same showtime	Error message shown	**Pass**
Book a broken seat	Error message shown	**Pass**
Book seat from a cancelled booking showtime	Seat available again	**Pass**
Book seat for a cancelled showtime	Error message shown	**Pass**

6.6 User Tests

Test	Expected Result	Pass/Fail
Add user with valid data	User added successfully	**Pass**
Add user with duplicate email	Error message shown	**Pass**
Add user with password under 6 characters	Error message shown	**Pass**
Deactivate a user	isActive set to 0, cannot log in	**Pass**

6.7 Seat Reserve Tests

Test	Expected Result	Pass/Fail
Add reservation for valid seat and showtime	Reservation added with auto expiry time	**Pass**
Add reservation with invalid showtime ID	Error message shown	**Pass**
Reservation auto expires after 15 minutes	Status changes to Expired automatically	**Pass**

6.8 Report Tests

Test	Expected Result	Pass/Fail
View all bookings report	All bookings shown with movie titles and screen types	**Pass**
Filter by Confirmed bookings	Only confirmed bookings shown, totals update	**Pass**
Filter by Cancelled bookings	Only cancelled bookings shown	**Pass**
Check total revenue calculation	Euro formatted total shown correctly	**Pass**

7. Validation Summary

The system implements extensive validation across all panels to ensure data integrity and provide clear feedback to users.

7.1 Input Validation

Validation	Details
Empty field check	All required fields must be filled before add or update
Email format	Must contain @ and . characters
Password length	Minimum 6 characters required
Date format	Must match YYYY-MM-DD pattern using regex
Datetime format	Must match YYYY-MM-DD HH:MM:SS pattern using regex
Duration validation	Must be a positive integer greater than 0
Amount validation	Must be a positive decimal greater than 0
Seat row format	Must be a single letter A-Z only
No changes detection	Update blocked if no fields have changed from current values

7.2 Other Validation

Rule	Details
Foreign key checks	All referenced IDs validated against database before insert or update
Duplicate prevention	Movies, emails, booking references, seats, showtimes, payment references and tickets all checked
Double booking prevention	Cannot book the same seat twice for the same showtime unless previous booking was cancelled
Broken seat check	Cannot book a seat where isWorking is false
Retired movie showtime	Cannot create a showtime for a movie with Inactive status
Retire movie with showtimes	Cannot retire a movie that has upcoming scheduled showtimes
Delete screen with seats	Cannot delete a screen that has seats assigned to it
Delete showtime with bookings	Cannot delete a showtime that has active bookings
Cancelled showtime booking	Cannot book seats for cancelled or completed showtimes
Auto expire reservations	Seat reserves automatically expire after 15 minutes via background timer